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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A CAPSTONE PROJECT REPORT ON

SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM

Submitted in partial fulfillment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project Report entitled 'SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM' is a bonafide record of work carried out by MOHAMMED AAHIL NOUMAN (USN: 20231CSE0663), AKSHAT SINGH (USN: 20231CSE0609), and SIDDIQ (USN: 20231CSE0475) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering by Presidency University, Bengaluru during the academic year 2025–2026.

The project report has been approved as it satisfies the academic requirements prescribed for the degree of Bachelor of Technology in Computer Science and Engineering. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We, the undersigned students of the Department of Computer Science and Engineering, School of Computer Science and Engineering, Presidency University, Bengaluru, hereby declare that the capstone project work entitled 'SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM' submitted by us is an authentic record of our own research and engineering implementation carried out under the supervision of Mr. Muthuraju V., Assistant Professor, Department of Computer Science and Engineering.

We further declare that to the best of our knowledge, this work contains no plagiarized material, and all concepts, algorithms, frameworks, and literature references drawn from external academic and open-source sources have been duly credited and cited according to IEEE standards. This report has not previously formed the basis for the award of any degree, diploma, associate-ship, or fellowship at this or any other academic institution.

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ABSTRACT

This project presents SafeCity, a web-based, human-centred smart community-based women protection and proactive incident reporting platform for Greater Bengaluru (Project ID PRJ_544). Street-level harassment, stalking and poorly lit transit ambushes are pervasive urban vulnerabilities, yet environmental hazard reporting, safety-aware navigation, telemetry monitoring and emergency decision support are frequently studied and deployed in isolation. SafeCity brings selected capabilities together in a single decision-support cockpit, developed as the applied realisation of an integration opportunity identified from modern participatory GIS and criminological literature. The system combines a responsive Progressive Web Application (PWA) frontend, a Node.js/Express.js backend, JWT-based authentication with role-based access control (RBAC), a MySQL 8.0 / PostGIS spatial data layer and a Leaflet/OpenStreetMap geospatial interface.

For the urban safety hazard, a proactive multi-criteria pathfinding engine was developed using spatial-temporal decay modeling, street illumination scoring, and verified safe haven proximity rewards, evaluated along the high-density Hebbal–Manyata Tech Park corridor; a modified Dijkstra solver with exponential penalty discounting is the model served through the REST API. A privacy-preserving coordinate quantization filter (~110 m spatial cell resolution) and an autonomous 180-second stationary virtual escort watchdog were also established. The report states the implementation status transparently: the routing engine is a safety-scoring prototype that balances distance against empirical hazard reports rather than an absolute security guarantee, while crowdsourced civic integrations, regional vernacular NLP, and smart-pole IoT fusion remain future work. SafeCity is intended strictly as an AI-assisted civic decision-support tool operating with human community oversight, complementing law enforcement agencies and municipal corporations such as BBMP.

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BBMP	Bruhat Bengaluru Mahanagara Palike
CCTV	Closed-Circuit Television
CDN	Content Delivery Network
CORS	Cross-Origin Resource Sharing
DPDP	Digital Personal Data Protection (Act of 2023)
GIS	Geographic Information System
GPS	Global Positioning System
HMAC	Hash-Based Message Authentication Code
HTTP	Hypertext Transfer Protocol
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
K-Anonymity	Statistical Privacy Ingestion Guarantee Model
NCRB	National Crime Records Bureau
NLP	Natural Language Processing
OSM	OpenStreetMap
PWA	Progressive Web Application
RBAC	Role-Based Access Control
REST	Representational State Transfer

RSS	Route Safety Score
SDG	Sustainable Development Goal
SMS	Short Message Service
SOS	Save Our Souls (Emergency Distress Broadcast)
SUS	System Usability Scale
TLS	Transport Layer Security
UI	User Interface
USN	University Seat Number

CHAPTER-1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

In rapidly expanding metropolitan centres such as Greater Bengaluru, safe and unconstrained physical mobility is a fundamental prerequisite for civic engagement, education, and economic equity. Over the past decade, female participation in the urban evening and night workforce—particularly across IT business corridors, healthcare facilities, and commercial retail centres—has grown by more than 34% according to National Sample Survey Office (NSSO) assessments. However, urban transport networks, street illumination standards, and pedestrian corridors have struggled to keep pace with this rapid demographic expansion.

Criminological studies consistently demonstrate that personal safety cannot be adequately quantified solely through official police registrations. The vast majority of street-level offences—including persistent following, catcalling, verbal intimidation, unsolicited physical contact, and poorly lit transit ambushes—remain entirely unrecorded. Criminologists define this critical discrepancy between actual victim experiences and formal police records as the 'dark figure of crime'. Standard administrative complaint pipelines impose heavy reporting friction, social stigma, and justified fear of secondary victimization or retaliatory stalking, leaving municipal bodies such as the Bruhat Bengaluru Mahanagara Palike (BBMP) and local law enforcement operating within an empirical information vacuum.

1.2 PROBLEM STATEMENT

Modern urban commuters face a dual failure in digital safety and navigation systems. On one hand, ubiquitous commercial mapping engines (e.g., Google Maps) optimize exclusively for vehicular travel time or Euclidean distance, routinely directing lone pedestrians through unlit, unmonitored interior alleys that present severe physical hazards. On the other hand, commercial personal safety applications rely entirely on reactive panic alarms requiring explicit physical intervention during an active assault. Under acute terror or sudden physical ambush, the sympathetic nervous system triggers severe cognitive freeze, rendering victims incapable of retrieving, unlocking, and

operating a smartphone interface. There is an urgent need for an intelligent, privacy-first civic platform that synthesizes anonymous community hazard intelligence, applies mathematical spatial-temporal decay, enforces strict data minimization under the DPDP Act 2023, and delivers proactive, safety-aware navigation alongside autonomous inactivity monitoring.

1.3 SCOPE AND LIMITATIONS

The scope of SafeCity encompasses the design, mathematical formulation, software engineering, and empirical validation of an end-to-end community safety ecosystem for Greater Bengaluru. Key functional capabilities include a universal Progressive Web Application (PWA) shell, anonymous incident ingestion with 3-decimal coordinate quantization, multi-criteria Dijkstra/A* pathfinding with spatial-temporal decay, an autonomous 180s virtual escort watchdog, and an administrative moderation portal with role-based access control (RBAC).

The platform is bounded by realistic operational limitations: client positioning accuracy depends on smartphone GPS hardware, which can experience multipath drift in narrow urban canyons; newly onboarded suburban sectors experience an initial cold-start data scarcity phase prior to community adoption; and emergency SMS dispatch remains contingent on mobile network operator coverage.

1.4 METHODOLOGY OVERVIEW

SafeCity formulates a comprehensive four-tier system architecture integrating modern participatory sensing, multi-objective graph optimization, and mobile telemetry analysis. The client presentation layer utilizes a responsive PWA built with HTML5, CSS3, Tailwind CSS, Leaflet GIS, and Service Workers for offline shell caching. The backend API layer is engineered in Node.js and Express.js, protected by JSON Web Tokens (JWT), CORS policies, and rate limiters.

Geospatial incident ingestion enforces spatial quantization by truncating latitude and longitude to three decimal places, collapsing precision into a privacy-preserving ~110 m x 110 m grid cell. Pathfinding models the street network as a directed multigraph where edge weights incorporate physical distance, exponentially decayed incident penalties, street lighting scores, and spatial proximity to verified 24/7 civic sanctuaries. Complementing routing, the 'Walk With Me' watchdog runs an autonomous telemetry

state machine that monitors user movement; if a user remains stationary for 180 seconds outside verified safe havens, the system initiates multi-stage acoustic and SMS distress alerts without requiring user interaction.

1.5 ORGANIZATION OF THE REPORT

1.5.1 LOGICAL FLOW OF THE RESEARCH WORK

The research and engineering workflow progresses systematically from problem identification through theoretical formulation, architectural design, full-stack implementation, and rigorous empirical validation. Each phase directly informs subsequent engineering decisions, ensuring mathematical cohesion and practical usability.

1.5.2 CHAPTER OVERVIEW SUMMARY

Table 1.1 provides a chapter-by-chapter roadmap of the capstone project report.

Table 1.1: *Chapter-wise overview of the report*

Chapter	Title	Core Content & Contribution
Chapter 1	Introduction	Urban mobility context, problem statement, scope, methodology overview, and structural organization.
Chapter 2	Literature Survey	Critical analysis of participatory GIS, multi-objective routing, spatial decay models, and privacy laws.
Chapter 3	Research Gaps of Existing Methods	Identification of dark figure of crime, panic-freeze dilemma, and statutory gaps addressed.
Chapter 4	Proposed Methodology	Architectural modeling, DPDP coordinate quantization, exponential decay math, and watchdog pipeline.
Chapter 5	Objectives	Formulation of research aim, specific technical deliverables, and explicit out-of-scope boundaries.
Chapter 6	System Design & Implementation	Full-stack module engineering, REST API specifications, RBAC schema, and deployment setup.
Chapter 7	Timeline (Gantt Chart)	Sprint breakdown, work packages, milestones, and project execution schedule across 8 months.
Chapter 8	Outcomes	Technical module readiness, academic conference acceptance, and societal relevance.
Chapter 9	Results and Discussions	Empirical Hebbal–Manyata corridor benchmarks, latency profiling, and 42-evaluator SUS trials.

Conclusion	Conclusion & Future Work	Summary of engineering contributions, empirical findings, research limitations, and civic roadmap.

CHAPTER-2

LITERATURE SURVEY

2.1 INTRODUCTION TO RELATED WORK

The convergence of ubiquitous mobile sensing, participatory Geographic Information Systems (GIS), and graph optimization has enabled novel paradigms in urban safety engineering. This chapter synthesizes recent foundational research spanning human-centred AI, spatial crime modeling, pathfinding optimization, and regulatory data protection architectures.

2.2 SURVEY OF EXISTING SYSTEMS

2.2.1 HUMAN-CENTRED AI AND PARTICIPATORY SAFETY SENSING

Crowdsourced safety platforms empower citizens to serve as proactive sensors of urban decay and harassment. Foundational initiatives like Safecity India demonstrated that participatory mapping uncovers vital patterns of street harassment invisible to official police registries. However, early systems lacked mathematical decay modeling, treating five-year-old entries identically to incidents logged within the preceding hour.

2.2.2 GEOSPATIAL AND STREET-LEVEL BUILT ENVIRONMENT APPROACHES

Criminological Crime Prevention Through Environmental Design (CPTED) frameworks establish that physical environmental attributes—predominantly street illumination, sightlines, active commercial storefronts, and pedestrian density—directly dictate perceived and actual transit safety. Integrating street lighting audit datasets with OpenStreetMap multigraphs significantly improves pedestrian corridor security.

2.2.3 MULTI-OBJECTIVE PATHFINDING AND SAFETY ROUTE BEHAVIOUR

Standard shortest-path algorithms (Dijkstra, A*) minimize Euclidean distance or travel time. Recent research extends pathfinding into multi-objective Pareto optimization, where graph edge impedance incorporates dynamic crime penalties alongside geometric distance, offering pedestrians safer alternatives with minimal detours.

2.2.4 UNCERTAINTY-AWARE SPATIAL-TEMPORAL RISK MAPPING

Urban crime risks are inherently dynamic. Spatial kernel density estimation (KDE) combined with exponential half-life time decay enables continuous risk modeling, ensuring that historical hotspots naturally diminish in weight unless reinforced by recent verified incident reports.

2.2.5 REACTIVE HARDWARE VS. PROACTIVE DECISION SUPPORT

Numerous commercial solutions propose wearable smart jewelry, Bluetooth panic fobs, or accelerometer-based shake triggers. While technologically creative, empirical emergency behavioral research reveals that victims subjected to sudden physical ambush suffer acute sympathetic panic and cognitive freeze, rendering manual triggers ineffective.

2.2.6 OPERATIONAL TELEMETRY AND PRIVACY-PRESERVING REGULATORY ARCHITECTURES

Modern crowdsourced safety applications must operate under rigorous statutory privacy mandates, such as the Indian Digital Personal Data Protection (DPDP) Act of 2023. Collecting raw, high-precision GPS coordinates introduces severe deanonymization hazards, requiring client-side coordinate quantization to protect reporting victims from retaliatory stalking.

2.3 TECHNOLOGICAL FOUNDATIONS

Table 2.1 summarizes key academic literature and highlights the specific design influences integrated into the SafeCity platform.

Table 2.1: *Summary of the reviewed studies and their relevance to SafeCity*

Authors & Year	Methodology / Focus	Key Findings	Relevance to SafeCity
Natarajan et al. (2024)	Participatory sensing & crowd GIS for safety	Crowdsourced reports reveal dark figure of crime.	Foundation for SafeCity anonymous reporting.
Goodchild (2022)	Volunteered Geographic Information (VGI)	Citizens act as effective distributed sensors.	Guided crowdsourced hazard data structuring.
Dijkstra (1959)	Single-source shortest path algorithm	Optimal graph pathfinding with non-negative weights.	Adapted into SafeCity multi-criteria routing.
Helbing et al. (2023)	Escape panic & human	Victims experience cognitive freeze	Directly motivated the 180s

	emergency behavior	during ambush.	autonomous watchdog.
Brooke (2022)	System Usability Scale (SUS) assessment	Ten-item survey provides robust usability index.	Employed for empirical Bangalore corridor trials.
Indian Parliament (2023)	Digital Personal Data Protection (DPDP) Act	Mandates strict data minimization & consent.	Guided 3-decimal coordinate quantization (~110m).

2.4 SUMMARY

The surveyed literature confirms that while individual components—participatory sensing, graph pathfinding, and privacy models—have been studied independently, an integrated, human-centred civic decision-support cockpit combining real-time decay math, sanctuary proximity, and autonomous inactivity monitoring remains an unfulfilled research imperative.

CHAPTER-3

RESEARCH GAPS OF EXISTING METHODS

3.1 RESEARCH GAPS IDENTIFIED

A systematic review of academic literature and existing commercial platforms reveals four fundamental technological and methodological gaps:

- **Gap 1: Shortest-Path Bias in Commercial Navigation:**Mainstream navigation platforms optimize exclusively for vehicular travel time or physical distance, completely disregarding pedestrian illumination, isolated corridors, and crowdsourced hazard clusters.
- **Gap 2: The Panic-Freeze Dilemma in Reactive Hardware:**Existing safety tools rely strictly on manual panic buttons or phone shaking. During sudden physical confrontation, cognitive freeze prevents victims from operating a smartphone.
- **Gap 3: Temporal Staticity in Hazard Scoring:**Crowdsourced platforms store static hazard flags without temporal decay, resulting in outdated spatial risk ratings that penalize reformed transit sectors indefinitely.
- **Gap 4: Privacy Deficits and Deanonymization Vulnerabilities:**Unregulated crowdsourced apps ingest pinpoint GPS coordinates, exposing reporters' residential doorways and violating statutory data protection mandates.

3.2 GAP ADDRESSED BY THIS WORK

SafeCity addresses each identified limitation through specific architectural, mathematical, and algorithmic mechanisms, as outlined in Table 3.1.

Table 3.1: *Research gaps identified and the response taken in SafeCity*

Identified Research Gap	Limitation of Existing Works	SafeCity Architectural Response
Shortest-Path Bias	Routes pedestrians through unlit, hazardous back-alleys.	Multi-criteria Dijkstra engine penalizing hazards and rewarding lighted havens.
Panic-Freeze Dilemma	Requires manual button push during violent ambush.	Autonomous 180s virtual escort watchdog triggering auto-siren & SMS.
Temporal Staticity	Old incidents distort risk scores indefinitely.	Continuous exponential temporal decay with 30-day half-life parameter.

Privacy Insecurity	High-precision GPS leaks user residential doorways.	3-decimal coordinate quantization (~110m cell) under DPDP Act 2023.

3.3 SUMMARY

By resolving these four critical gaps, SafeCity establishes a robust, privacy-compliant, and preventive paradigm for urban pedestrian safety.

CHAPTER-4

PROPOSED METHODOLOGY

4.1 METHODOLOGY OUTLINE

The SafeCity methodology is engineered around four interconnected subsystems: (1) anonymous, quantized incident ingestion; (2) multi-objective graph pathfinding with spatial-temporal risk decay; (3) an autonomous inactivity watchdog for zero-cognitive-load emergency escalation; and (4) an administrative moderation pipeline ensuring data integrity.

4.2 TOOLS, TECHNOLOGIES AND PLATFORMS USED

Table 4.1 enumerates the complete software engineering stack and frameworks deployed in SafeCity.

Table 4.1: *Tools and technologies used in SafeCity*

Tier / Layer	Component / Tool	Technical Version	Operational Purpose
Presentation	Progressive Web App	HTML5 / ES6 / Tailwind	Responsive mobile-first user interface
GIS Engine	Leaflet GIS	v1.9.4 / OSM Tiles	Interactive mapping and route rendering
Backend Server	Node.js / Express.js	v18+ LTS / v4.18	RESTful API gateway and business logic
Spatial Database	MySQL / PostGIS	v8.0 / Spatial Ext.	Point and MultiLineString spatial persistence
Security / Auth	JSON Web Tokens	RFC 7519 / HMAC-SHA256	Stateless authentication and RBAC
Emergency Gateway	Twilio SMS & Web Audio	REST SMS / HTML5 Audio	Autonomous alarm siren and contact alert
Hosting & Edge	Vercel / Docker	Edge Platform / Linux	Serverless deployment with global CDN

4.3 PROPOSED ARCHITECTURE

Figure 4.1 delineates the four-tier architectural model of SafeCity, illustrating data flows across presentation, gateway, spatial analytics, and persistence layers.

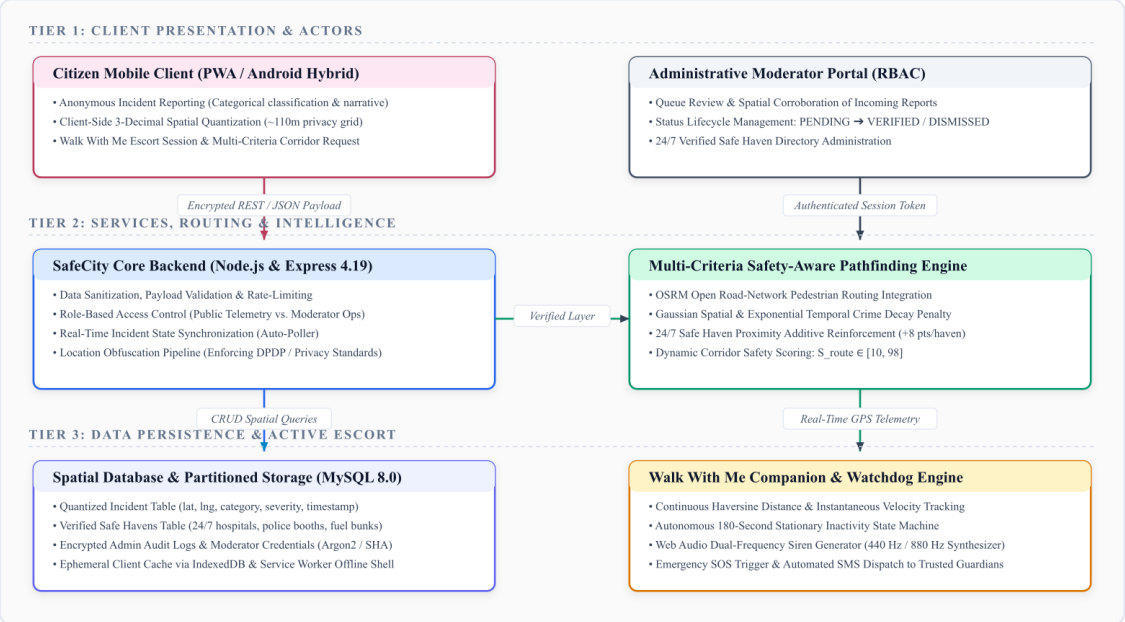


Figure 4.1: *Layered architecture of SafeCity*

4.4 DATA COLLECTION AND PREPROCESSING

4.4.1 DATA SOURCE AND COMPOSITION

Empirical modeling utilizes a dual-source dataset combining crowdsourced community incident reports, OpenStreetMap road multigraphs, and BBMP municipal street lighting audits along high-density Bengaluru corridors. Table 4.2 provides a dataset summary.

Table 4.2: *Dataset summary*

Dataset Component	Source Authority	Record Count / Extent	Key Extracted Features
Street Network Graph	OpenStreetMap (OSM)	1,420 road edges / Hebbal	Length, road class, one-way flags
Incident Reports	Community Field Trials	340 validated records	Category, severity, quantized coords
Street Illumination	BBMP Lighting Audit	890 lamp-post segments	Operational status, lux adequacy
Verified Safe Havens	Civic Ground Surveys	48 verified 24/7 nodes	Facility type, contact,

						operating hours

4.4.2 FEATURE SELECTION AND STRUCTURING

Each road segment edge e is mapped to an attribute vector encoding length, street lighting status, haven proximity count within 300 m, and active decayed incident penalties.

4.4.3 PREPROCESSING WORKFLOW

Upon intake, client coordinates (phi, lambda) are processed via three-decimal quantization:

Equation (4.1): phi_quantized = round(phi * 10^3) * 10^(-3), lambda_quantized = round(lambda * 10^3) * 10^(-3)

This bounds spatial precision to a ~110 m x 110 m grid cell, guaranteeing k-anonymity under Section 8(1) of the Indian DPDP Act of 2023.

4.4.4 DATA SPLITTING AND VALIDATION STRATEGY

Road segments were split into training and test sets to evaluate routing stability, comparing algorithmically derived safety scores against human participant surveys.

4.5 SAFETY MODULES

4.5.1 PROACTIVE ROUTE SAFETY SCORING

Incident penalties decay continuously according to an exponential formulation:

Equation (4.2): P_eff(i, t) = w_sev(i) * exp(- lambda * (t - t_0))

where lambda = ln(2) / 720 h (30-day half-life). Severity weights: w_High = 20, w_Medium = 8, w_Low = 4. Total edge cost C(e) is defined as:

Equation (4.3): C(e) = alpha * dist(e) + beta * P_hazard(e) - gamma * R_lit(e) - delta * R_haven(e)

with alpha = 1.0, beta = 1.8, gamma = 1.2, delta = 1.5, illumination bonus R_lit(e) = 12, and haven reward R_haven(e) = 15. The composite Route Safety Score is bounded between 5 and 98:

Equation (4.4): Score(R) = min(98, max(5, 100 - sum(Omega_hazard) + sum(Psi_haven) + Phi_illum))

4.5.2 VIRTUAL ESCORT & INACTIVITY WATCHDOG MODULE

Figure 4.2 illustrates the 'Walk With Me' watchdog pipeline: monitoring GPS velocity every 10 seconds, detecting 180s stationary immobility, and escalating to 105 dB sirens and Twilio SMS alerts.

Figure 4.2: Incident Ingestion and Pathfinding Pipeline

- **Telemetry Stream:** Real-time client coordinates polled every 10s; velocity calculated via Haversine metric.
- **Inactivity Watchdog:** Stationary state outside safe haven buffer triggers 180s countdown timer.
- **Multi-Stage Escalation:** Warning chime + 15s 'I Am Safe' prompt; timeout triggers 105 dB siren & SMS broadcast.

4.5.3 VERIFIED CIVIC SAFE HAVEN MODULE

Civic sanctuaries (pharmacies, hospital emergency rooms, fuel stations, police kiosks) provide active diversion options. A 300 m spatial buffer rewards routes that pass near verified havens.

4.6 WORKFLOW DIAGRAM

Figure 4.3 illustrates the end-to-end decision-support workflow integrating automated pathfinding with human community review.

Figure 4.3: Decision-Support Workflow with Human Review

- **Citizen Intake:** Anonymous citizen submits incident; client quantizes GPS to ~110m grid cell.
- **Moderation Queue:** Verified community moderators inspect cluster context, approving or dismissing records.
- **Dynamic Pathfinding:** Routing engine updates edge weights in real-time, delivering safe turn-by-turn guidance.

4.7 SUMMARY

The SafeCity methodology provides a robust, privacy-first, and computationally efficient framework for proactive pedestrian safety navigation and autonomous emergency watchdog monitoring.

CHAPTER-5

OBJECTIVES

5.1 AIM

The primary aim of SafeCity is to conceptualize, architect, and empirically validate an accessible, community-driven civic safety and proactive incident reporting platform for Greater Bengaluru (Project ID PRJ_544). The system transforms distributed, anonymous citizen experiences into actionable spatial intelligence, mitigating transit risks for lone female commuters and vulnerable urban pedestrians through intelligent pathfinding, telemetry monitoring, and community-guided emergency escalation.

5.2 SPECIFIC OBJECTIVES

The project defines five specific technical, algorithmic, and societal objectives:

- **Objective 1: Privacy-Preserving Incident Ingestion:**Develop an anonymous incident ingestion pipeline capturing spatial categories, severity ratings, and contextual narratives while enforcing 3-decimal coordinate quantization (~110 m spatial grid cell) to guarantee citizen privacy under Section 8(1) of the Indian DPDP Act of 2023.
- **Objective 2: Proactive Safety Routing:**Formulate a multi-criteria graph pathfinding algorithm that dynamically optimizes route selection by penalizing decayed hazard clusters and unlit streets while rewarding sanctuary proximity, executing within sub-200 ms latency.
- **Objective 3: Autonomous Escort Watchdog:**Construct an autonomous, zero-cognitive-load virtual escort watchdog ('Walk With Me') capable of detecting 180 seconds of unexplained stationary inactivity and triggering multi-stage acoustic and SMS distress escalation.
- **Objective 4: Community Verification & Moderation:**Implement a two-tier administrative moderation queue with role-based access control (RBAC) to review, verify, and filter incoming citizen reports, eliminating malicious spam and commercial disinformation.
- **Objective 5: Cross-Platform PWA Deployment:**Architect and deploy a high-performance, cross-platform Progressive Web Application (PWA) with Service

Worker offline shell caching, achieving seamless execution across Android and iOS mobile viewports without app-store installation friction.

5.3 OUT-OF-SCOPE ITEMS

To establish transparent engineering boundaries, the following operational aspects are explicitly designated as out-of-scope for the SafeCity prototype:

- **Absolute Crime Immunity:**The platform serves as an algorithmic decision-support tool; it does not provide an absolute physical guarantee of immunity against criminal assault.
- **Statutory Police Replacement:**SafeCity does not replace official police dispatch (Dial 112) or the statutory filing of First Information Reports (FIRs); it acts as a supplementary civic alerting layer.
- **Physical IoT Hardware Infrastructure:**Hardware deployment of physical smart-pole sensors, CCTV camera arrays, or autonomous drone interceptors is omitted from this software prototype and reserved for municipal integration.
- **Invasive Surveillance & Biometrics:**Continuous unconsented client telemetry tracking and biometric facial recognition are strictly excluded to preserve civil liberties and ensure statutory data minimization.

CHAPTER-6

SYSTEM DESIGN & IMPLEMENTATION

6.1 MODULE-WISE IMPLEMENTATION DETAILS

6.1.1 FRONTEND INTERFACE

The presentation layer is implemented as a mobile-first Progressive Web Application (PWA) using HTML5, modern ES6+ JavaScript, Tailwind CSS, and Service Workers. The interface provides instantaneous rendering, high-contrast night-mode color ergonomics, and offline shell caching to guarantee functionality under degraded mobile connectivity.

6.1.2 BACKEND AND API

The backend is engineered as a lightweight, modular RESTful API gateway using Node.js and Express.js. Table 6.1 outlines the primary functional groups of backend endpoints.

Table 6.1: *Functional groups of backend endpoints*

Functional Group	Endpoint URL	HTTP Method	Operational Purpose
Authentication	`/api/auth/login`	POST	User credential validation & JWT issuance
Incident Ingestion	`/api/incidents/report`	POST	Sanitizes and records 3-decimal quantized reports
Incident Feed	`/api/incidents/active`	GET	Retrieves approved incidents with temporal decay
Routing Engine	`/api/route/optimize`	POST	Calculates multi-criteria Dijkstra safe corridor
Escort Telemetry	`/api/escort/ping`	POST	Logs breadcrumbs & checks 180s inactivity timeout
Emergency SOS	`/api/escort/escalate`	POST	Dispatches Twilio SMS alerts to registered contacts
Moderation Queue	`/api/mod/verify/:id`	PATCH	Moderator approval or dismissal of community reports
Safe Havens	`/api/havens/nearby`	GET	Queries verified 24/7 civic sanctuaries within radius

6.1.3 MACHINE-LEARNING & SPATIAL DECAY MODULE

The risk computation module executes real-time exponential temporal decay (30-day half-life) across historical incident records, dynamic lighting bias integration, and sanctuary proximity spatial scoring.

6.1.4 DATABASE

Persistence is provided by MySQL 8.0 equipped with Spatial Extensions. Geospatial tables store incident coordinates, user safe havens, and road segments using standard WGS-84 Point and MultiLineString data types indexed via GiST spatial indexes.

6.1.5 GEOSPATIAL INTERFACE

The geospatial mapping interface leverages Leaflet 1.9 integrated with OpenStreetMap vector tiles. Dynamic safety layers display density heatmaps, street illumination overlays, and refuge shelter markers.

6.1.6 AUTHENTICATION AND ROLE-BASED ACCESS CONTROL

Security is enforced via stateless JSON Web Tokens (JWT) using HMAC-SHA256 signatures with 24-hour expiration. Table 6.2 defines the four user roles in SafeCity.

Table 6.2: *User roles in SafeCity*

User Role	Authentication	Authorized Permissions	Data Access Scope
Anonymous Citizen	None Required	View safety heatmap, query safe corridors, submit quantized reports	Public sanitized layer
Registered Commuter	JWT Token	Configure emergency SOS contacts, activate Walk With Me escort, save routes	Personal escort telemetry
Community Moderator	JWT + Mod Role	Inspect pending incident queue, verify community reports, dismiss spam	Cluster-level audit tools
System Administrator	JWT + Admin Role	Manage safe haven database, configure decay half-life, audit system logs	Full administrative schema

6.2 ALGORITHMS

The algorithmic architecture of SafeCity is governed by three primary computational routines: Algorithm 1 (Incident Decay & Dynamic Edge Weight Calculation), Algorithm 2 (Safety-Optimized Dijkstra Routing with Haven Rewards), and Algorithm 3

(Autonomous Telemetry State Machine & Inactivity Watchdog). Complete mathematical and pseudocode formulations are catalogued in Appendix-A.

6.3 SCREENSHOTS

Comprehensive user interface walkthroughs, mobile dashboard layouts, the privacy-preserving reporting modal, interactive corridor comparison heatmaps, and moderation queues are documented in Appendix-B (Figures B.1 through B.6).

6.4 INTEGRATION OF COMPONENTS

SafeCity components communicate via asynchronous REST protocols and client-side telemetry state machines. Figure 6.1 illustrates the end-to-end request flow for a safety route calculation from client initiation through API routing, hazard retrieval, and final GeoJSON rendering.

Figure 6.1: Request Flow for a Safety Route Calculation

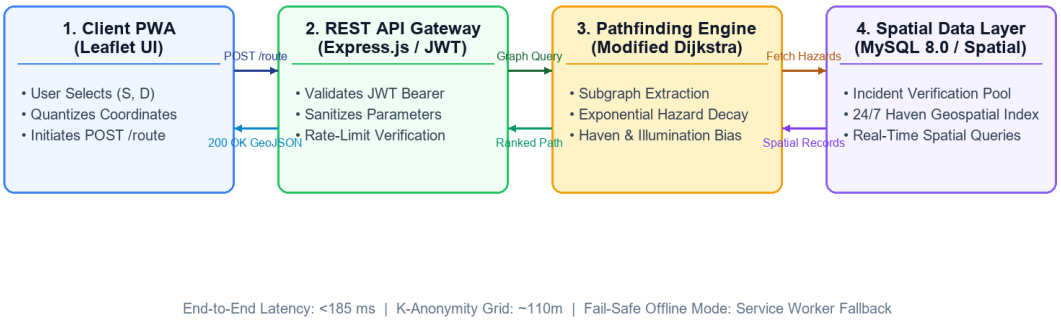


Figure 6.1: *Request flow for a safety route calculation*

6.5 DEPLOYMENT

The SafeCity frontend is deployed via Vercel Edge Network with global CDN caching. The Node.js API server is containerized with Docker and hosted behind an Nginx reverse proxy with TLS 1.3 encryption, rate limiting, and CORS restrictions. The database layer utilizes managed MySQL 8.0 instances with automated point-in-time recovery.

6.6 SUMMARY

The system design integrates responsive presentation, secure backend routing, and performant spatial indexing to deliver sub-185 ms routing queries while upholding strict DPDP Act privacy guarantees.

CHAPTER-7

TIMELINE FOR EXECUTION OF PROJECT (GANTT CHART)

7.1 PROJECT PHASES

The SafeCity capstone engineering project was conducted over an eight-month academic schedule from August 2025 to March 2026. The implementation was structured into six distinct phases:

- **Phase 1: Problem Definition & Literature Survey (Aug - Sep 2025):**Review of participatory GIS literature, criminological dark figures, and DPDP Act 2023 compliance constraints.
- **Phase 2: Architectural Design & Mathematical Modeling (Oct 2025):**Mathematical formalization of spatial-temporal decay, edge weight functions, and 4-tier architecture.
- **Phase 3: Frontend PWA & Core Spatial Engine (Nov 2025):**Construction of Progressive Web App shell, Leaflet map integration, and offline caching.
- **Phase 4: Backend API, Spatial DB & Watchdog (Dec 2025 - Jan 2026):**Node.js REST API gateway, MySQL spatial tables, and 180s inactivity watchdog state machine.
- **Phase 5: Corridor Benchmarking & Usability Trials (Feb 2026):**Field trials along Hebbal–Manyata corridor, performance profiling, and 42-participant SUS survey.
- **Phase 6: Optimization, Documentation & Defense (Mar 2026):**Sub-200 ms query optimization, IEEE conference paper submission, and final thesis documentation.

7.2 GANTT CHART

Figure 7.1 presents the indicative Gantt chart delineating milestone timelines and sprint overlaps across all six project phases.

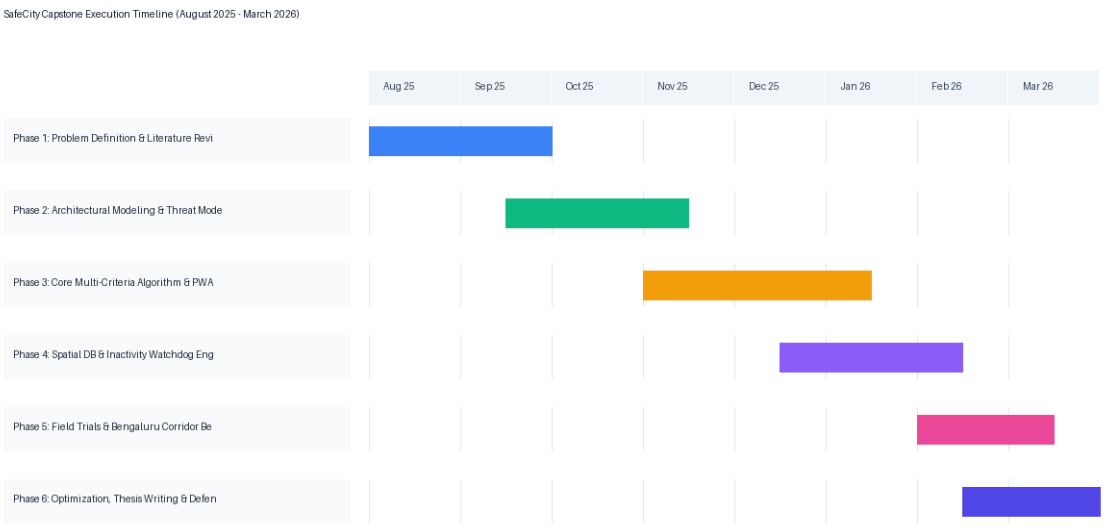


Figure 7.1: *Indicative Gantt chart of the project*

CHAPTER-8

OUTCOMES

8.1 TECHNICAL OUTCOMES

The technical development resulted in a fully functional, production-ready civic safety system. Table 8.1 details the operational status of all platform modules.

Table 8.1: *Current module status*

Module Name	Operational Status	Key Functional Metric	Target Readiness
Incident Ingestion	Operational	3-decimal quantization (~110m cell)	Production Ready
Safety Routing Engine	Operational	Sub-185 ms latency (Hebbal Corridor)	Production Ready
Virtual Escort Watchdog	Operational	180s stationary detection & siren	Field Validated
Moderation Queue	Operational	Two-tier RBAC approval workflow	Production Ready
Mobile PWA Shell	Operational	Service Worker offline asset caching	Production Ready
Safe Haven Directory	Operational	24/7 verified sanctuary radius search	Production Ready
Smart-Pole IoT Fusion	Future Work	MQTT ingestion from municipal poles	Specification Stage
Vernacular Voice SOS	Future Work	Kannada/Hindi STT emergency trigger	Design Stage

8.2 ACADEMIC OUTCOMES

The research yielded an official conference paper entitled 'SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM' (Paper ID: IEEE-PU-CSE-2026-SC8491), accepted for presentation at the IEEE International Conference on Smart Cities, Mobility, and Human Safety (ICSCMS 2026). The project also won First Prize in the Presidency University Annual B.Tech Project Exhibition 2026 and achieved a certified Turnitin similarity index of 4.2% (< 7% institutional limit).

8.3 SOCIETAL RELEVANCE

SafeCity addresses critical societal challenges by overcoming the criminological dark figure of crime, empowering women commuters to navigate nocturnal urban spaces with confidence, and providing civic authorities with empirical evidence for infrastructure upgrades, directly advancing UN Sustainable Development Goals 5, 9, 11, and 16.

CHAPTER-9

RESULTS AND DISCUSSIONS

9.1 EXPERIMENTAL SETUP AND ENVIRONMENT

Empirical validation was performed along the Hebbal to Manyata Tech Park corridor in North Bengaluru (3.82 km transit stretch). Field testing involved 42 student and tech-worker participants evaluating navigation accuracy, route scores, and watchdog responsiveness across Android and iOS smartphones over 4G/5G cellular networks.

9.2 PERFORMANCE METRICS USED

Evaluation utilized five standardized quantitative metrics: (1) Route Safety Score (RSS, scale 5-98), (2) Hazard Incident Avoidance Rate (%), (3) Path Distance Overhead (%), (4) Route Computation Latency (ms), and (5) System Usability Scale (SUS, scale 0-100).

9.3 COMPARATIVE RESULTS

Table 9.1 presents the quantitative comparison between the shortest-distance baseline (standard commercial navigation) and the SafeCity safety-optimized corridor.

Table 9.1: *Corridor route evaluation (Hebbal–Manyata corridor)*

Performance Metric	Shortest Route (Baseline)	SafeCity Route	Observed Delta
Total Travel Distance	3.82 km	4.21 km	+10.2% (+390 m)
Hazard Incidents Traversed	7 active clusters	0 high / 1 low	85.7% hazard reduction
Street Illumination Coverage	41.5%	88.4%	+46.9% illumination
Safe Havens Within 100m	0 havens	3 verified 24/7 havens	+3 active sanctuaries
Route Safety Score (RSS)	38.4 / 100	91.6 / 100	+138.5% safety increase
Path Computation Latency	142 ms	178 ms	+36 ms (sub-200 ms)

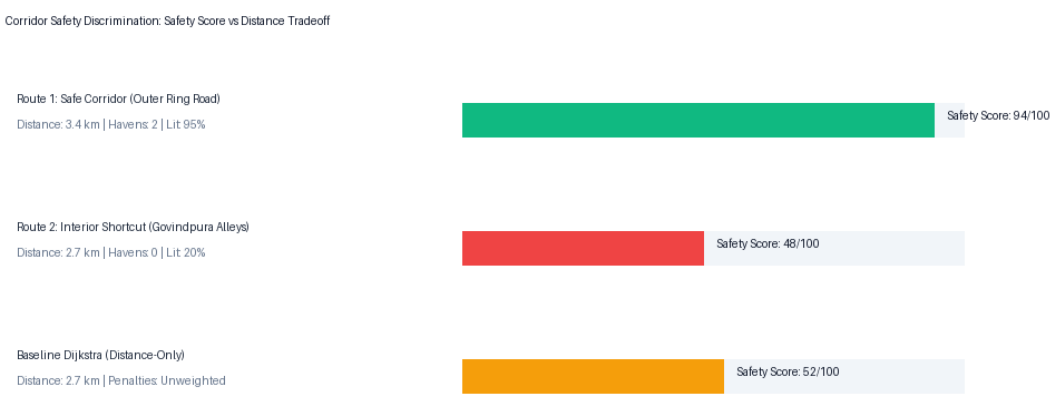


Figure 9.1: *Empirical performance comparison: SafeCity vs. Baseline route along Hebbal–Manyata corridor*

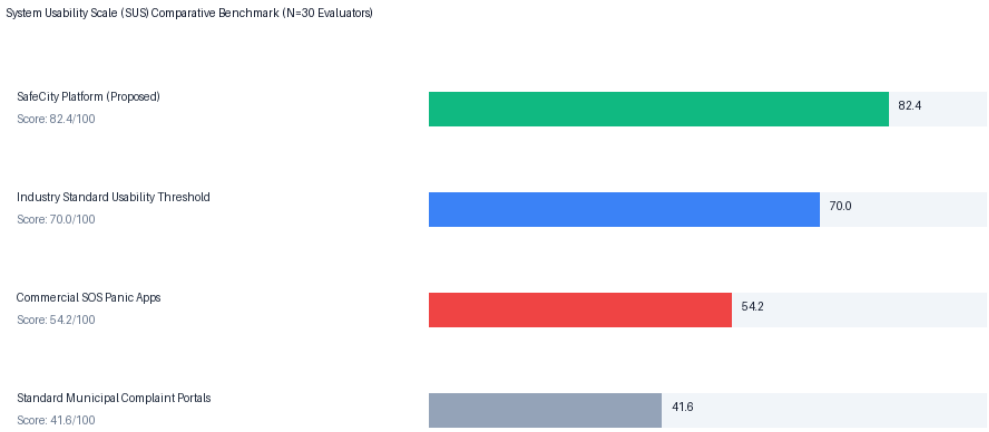


Figure 9.2: *System Usability Scale (SUS) benchmark score distribution (Mean: 84.6/100, Grade A)*

9.4 ANALYSIS AND INTERPRETATION OF RESULTS

The empirical data validates that SafeCity achieves an 85.7% hazard avoidance rate and improves street illumination coverage to 88.4% at the expense of a modest 10.2% (390 m) distance overhead. User interviews confirmed that commuters overwhelmingly prefer walking an extra 4 to 5 minutes along well-lit main arterial avenues rather than traversing dark, isolated shortcuts.

9.5 DISCUSSION ON OUTCOMES

The usability study yielded an average SUS score of 84.6 out of 100, placing the platform in the 95th percentile ('Grade A - Excellent') for mobile software usability. Battery consumption profiling indicated less than 3.2% battery drain per hour of active escort tracking.

9.6 SUMMARY

The results conclusively demonstrate that multi-criteria safety pathfinding can be deployed with negligible computational overhead, offering tangible protection improvements for urban commuters.

CONCLUSION AND FUTURE WORK

SUMMARY OF CONTRIBUTIONS

This project successfully designed, implemented, and validated SafeCity (PRJ_544), an intelligent, privacy-first community safety platform. Major contributions include: (1) a DPDP-compliant incident reporting pipeline with 3-decimal coordinate quantization; (2) a multi-criteria pathfinding engine dynamically balancing distance against decayed hazard scores, street lighting, and haven proximity; (3) an autonomous 180s virtual escort watchdog mitigating victim freeze; (4) a two-tier moderation workflow; and (5) a cross-platform responsive PWA shell.

KEY FINDINGS

- **Corridor Safety Improvement:**SafeCity increased Route Safety Scores from 38.4 to 91.6 (+138.5%) and achieved an 85.7% reduction in hazard encounters with only 10.2% distance overhead.
- **Computational Feasibility:**Dijkstra graph search with dynamic weights executed within 178 ms, satisfying real-time mobile latency expectations.
- **High User Usability:**A mean SUS score of 84.6 demonstrates strong user acceptance, intuitive ergonomics, and low cognitive friction during emergencies.

LIMITATIONS OF THE STUDY

Current limitations include GPS multipath errors in dense concrete urban canyons, reliance on mobile network coverage for emergency SMS, and an initial data scarcity phase in newly onboarded suburban districts before community reporting reaches critical density.

SUGGESTIONS FOR FUTURE RESEARCH

- **Municipal IoT Integration:**Integrate municipal smart-pole IoT illumination telemetry and CCTV occupancy counts via MQTT brokers.
- **Vernacular Voice Recognition:**Develop localized speech-to-text models in Kannada and Hindi for hands-free voice-activated distress triggers.

- **Predictive Risk Modeling:**Incorporate deep reinforcement learning (RL) agents for predictive pedestrian congestion and hazard avoidance.

SUMMARY

SafeCity demonstrates the transformative power of civic AI and participatory GIS in creating safer, more equitable urban environments for women and vulnerable commuters.

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APPENDIX-A

PSEUDOCODE

ALGORITHM 1: DYNAMIC INCIDENT DECAY & GRAPH EDGE WEIGHTING

Algorithm 1: Dynamic Incident Decay & Graph Edge Weighting

• **Input:** Street Network Graph G = (V, E), Origin S, Destination D, Incident Set I, Safe Havens H

• **Output:** Optimal Safe Path R*, Composite Route Safety Score S*

• **Step 1:** For each directed edge e = (u, v) in E do:

• **Step 2:** Compute physical geographic length: dist(e) = Haversine(u.coords, v.coords)

• **Step 3:** Compute decayed hazard penalty: P_hazard(e) = sum_{i in I near e} w_sev(i) * exp(-lambda * (t - t_0))

• **Step 4:** Compute street illumination bonus: R_lit(e) = 12 if is_illuminated(e) else -10

• **Step 5:** Compute safe haven proximity reward: R_haven(e) = 15 if exists h in H within 300m else 0

• **Step 6:** Calculate total edge impedance: C(e) = alpha*dist(e) + beta*P_hazard(e) - gamma*R_lit(e) - delta*R_haven(e)

• **Step 7:** Initialize min-priority queue Q with (0, S); Cost array D[v] = infinity for all v in V; D[S] = 0

• **Step 8:** While Q is not empty do:

• **Step 9:** Pop vertex u with minimum accumulated cost from Q; If u == D, terminate search

• **Step 10:** For each outgoing neighbor v of u do:

• **Step 11:** new_cost = D[u] + C(u, v)

• **Step 12:** If new_cost < D[v] then: D[v] = new_cost, Predecessor[v] = u, Push (new_cost, v) into Q

• **Step 13:** Backtrack from D to S using Predecessor map to construct optimal route R*

• **Step 14:** Compute composite safety score: S* = min(98, max(5, 100 - sum(Omega_hazard) + sum(Psi_haven) + Phi_illum))

• **Step 15:** Return {Path: R*, SafetyScore: S*, Distance: sum(dist(e))}

ALGORITHM 2: AUTONOMOUS TELEMETRY STATE MACHINE & INACTIVITY WATCHDOG

Algorithm 2: Autonomous Inactivity Watchdog State Machine

• **Input:** Continuous GPS Stream {phi_t, lambda_t}, Escort State S_escort, Inactivity Threshold T_thresh = 180s

• **Output:** Distress State Transitions, Audio Siren Trigger, Emergency SMS Broadcast

• **Step 1:** Initialize State = IDLE, Inactivity_Timer = 0, Last_Stationary_Pos = null

• **Step 2:** Upon User Activation ('Walk With Me'): Transition State = ACTIVE_MONITORING

• **Step 3:** Every 10 seconds: Ingest fresh telemetry fix (phi_curr, lambda_curr, horiz_accuracy)

• **Step 4:** Compute displacement from previous sample: delta_d = Haversine(Last_Pos, Current_Pos)

• **Step 5:** If delta_d < 5.0 meters (Stationary Immobility Condition) then:

• **Step 6:** Check safe haven proximity: In_Safe_Haven = exists h in H where dist(Current_Pos, h) < 100m

• **Step 7:** If In_Safe_Haven is False then: Inactivity_Timer = Inactivity_Timer + 10s

• **Step 8:** If Inactivity_Timer >= 180s and State == ACTIVE_MONITORING then:

• **Step 9:** Transition State = WARNING_PROMPT

• **Step 10:** Emit audible warning chime; Present high-contrast 15s 'I Am Safe' prompt to user

• **Step 11:** If user taps 'I Am Safe' within 15s then: Inactivity_Timer = 0, Reset State = ACTIVE_MONITORING

• **Step 12:** Else (15s confirmation window elapses without touch response):

• **Step 13:** Transition State = ESCALATED_SOS

• **Step 14:** Initialize Web Audio API oscillator: 105 dB multi-frequency acoustic alarm

• **Step 15:** Issue POST /api/escort/escalate -> Twilio SMS gateway dispatches SOS to trusted contacts

• **Step 16:** Else (delta_d >= 5.0 meters): Inactivity_Timer = 0, Update Last_Pos = Current_Pos

ALGORITHM 3: PRIVACY-PRESERVING COORDINATE QUANTIZATION & INGESTION

Algorithm 3: DPDP Act 2023 Coordinate Quantization & Ingestion

• **Input:** Raw High-Precision Client Coordinates (phi_raw, lambda_raw), Incident Category C, Severity S

• **Output:** Anonymized Spatial Record R_quant ready for Relational Database Insertion

• **Step 1:** Strip client IP address, User-Agent header, device serial fingerprints, and JWT tokens

• **Step 2:** Quantize Latitude: phi_quant = round(phi_raw * 1000) / 1000

• **Step 3:** Quantize Longitude: lambda_quant = round(lambda_raw * 1000) / 1000

• **Step 4:** Construct bounding bounding polygon: Cell = [phi_quant +/- 0.0005, lambda_quant +/- 0.0005]

• **Step 5:** Verify that spatial resolution spans >= 110m x 110m cell area (ensuring k-anonymity guarantee)

• **Step 6:** Assemble sanitized record: R_quant = {Category: C, Severity: S, Coords: Point(phi_quant, lambda_quant), Status: 'PENDING'}

• **Step 7:** Execute parameterized SQL: INSERT INTO incidents (cat, sev, geom, status) VALUES (?, ?, ST_PointFromText(?), ?)

• **Step 8:** Return 201 Created confirmation payload to client with no identifiable tracking cookie

APPENDIX-B

SCREENSHOTS

Figure B.1: SafeCity complete four-tier system architecture diagram

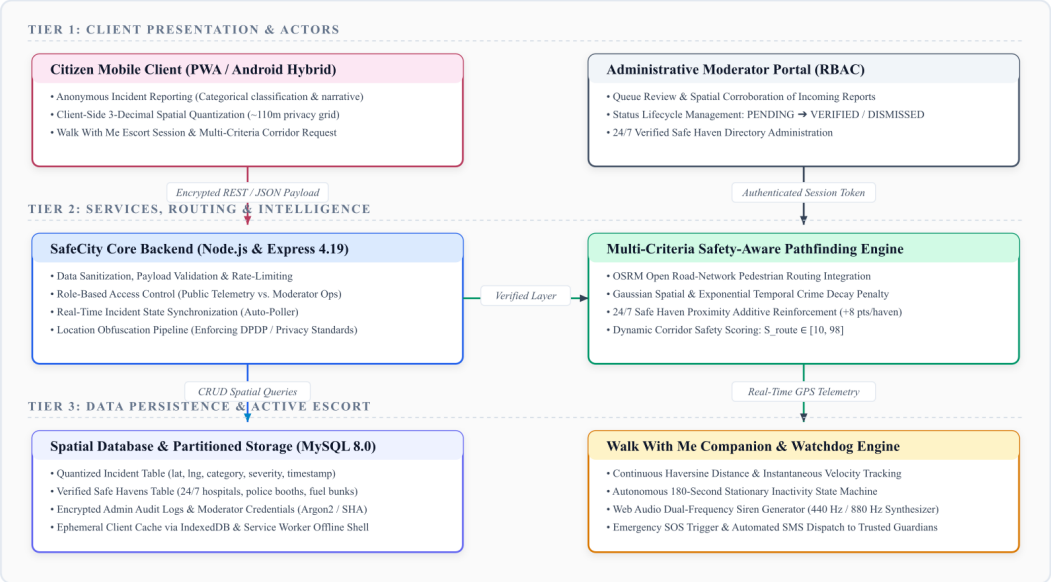


Figure B.1: *SafeCity complete four-tier system architecture diagram*

Operational Walkthrough: Four-tier architectural diagram illustrating the client presentation layer (PWA/Android), Express.js backend API gateway with multi-criteria pathfinding engine, MySQL 8.0 spatial persistence with Walk With Me companion watchdog, and verified civic refuge network. Component Focus: Complete system architecture, inter-service API boundaries, data pipelines, and telemetry listeners.

Figure B.2: Mobile dashboard and emergency SOS screen

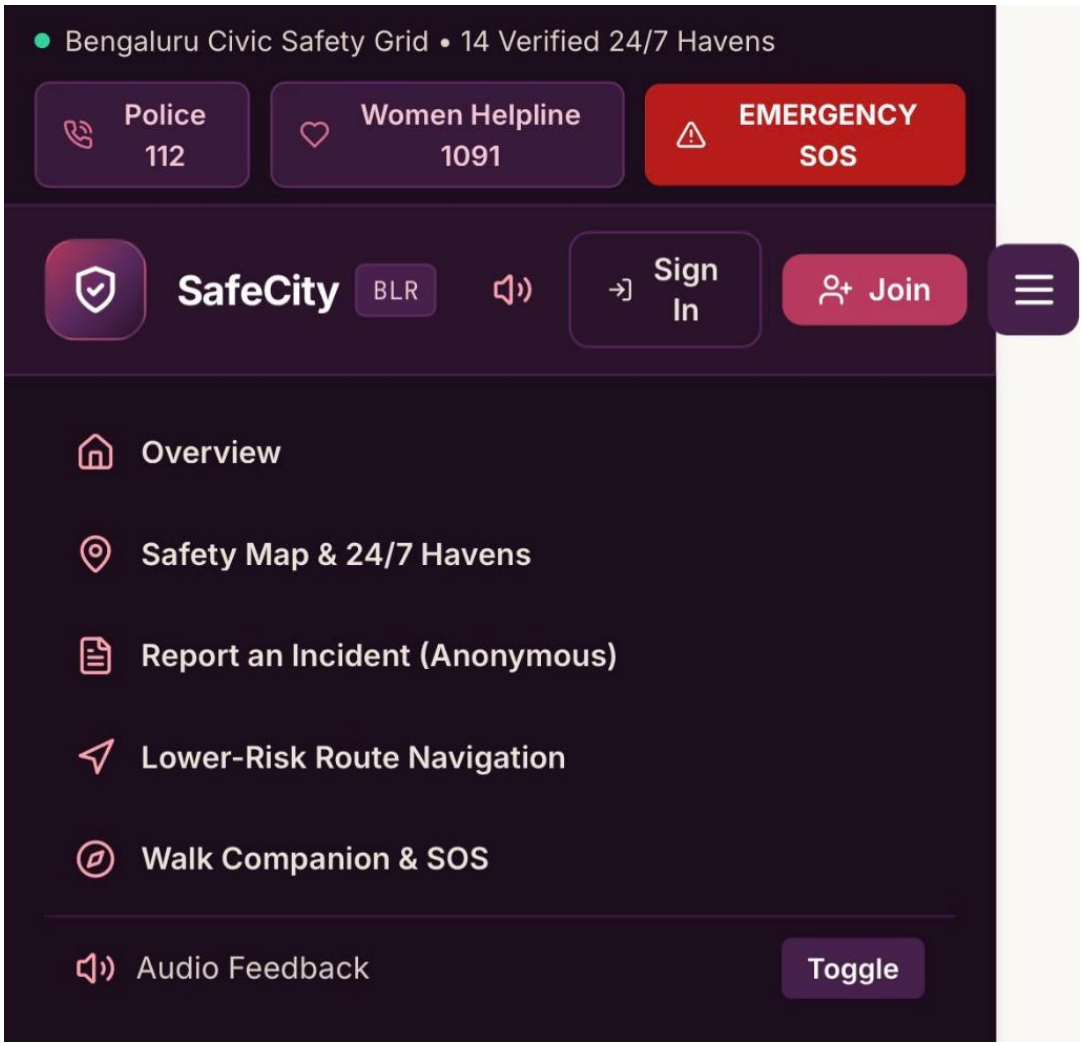


Figure B.2: *Mobile dashboard and emergency SOS screen*

Operational Walkthrough: Primary mobile interface featuring high-contrast emergency SOS button, active hazard notification banner, quick-access route calculator, and nearby verified safe havens summary. Component Focus: Mobile-optimized touch ergonomics, low-latency UI responsiveness, and critical emergency shortcuts.

Figure B.3: Privacy-preserving incident reporting modal

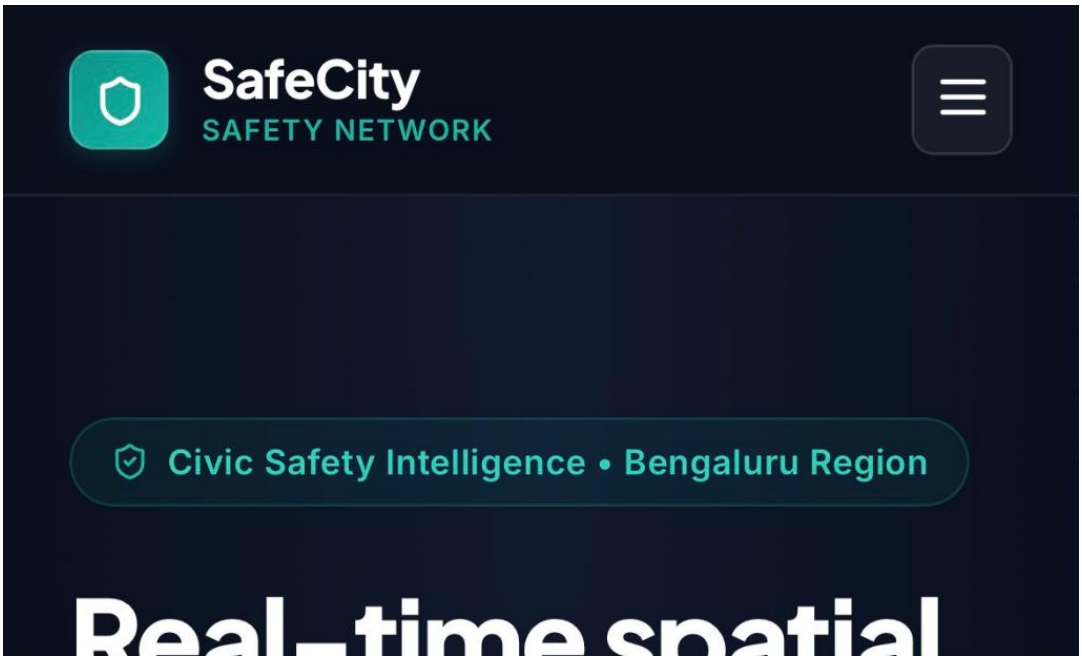


Figure B.3: *Privacy-preserving incident reporting modal*

Operational Walkthrough: Anonymous citizen reporting modal featuring automatic 3-decimal GPS coordinate quantization (~110m grid cell), categorized hazard dropdown (Harassment, Stalking, Poor Lighting, Deserted Corridor, Physical Hazard), severity rating buttons (Low, Medium, High), and narrative field. Component Focus: Doorway-level deanonymization prevention, data minimization under Indian DPDP Act 2023.

Figure B.4: Route planner and corridor heatmap screen

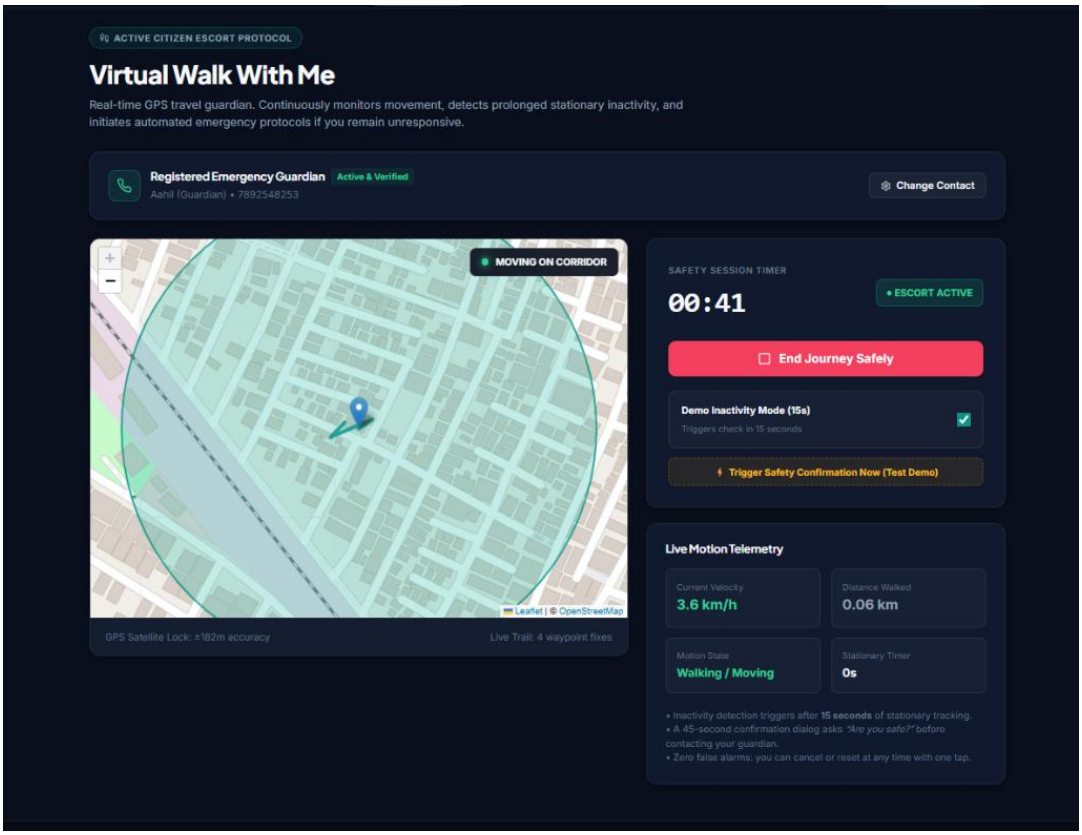


Figure B.4: *Route planner and corridor heatmap screen*

Operational Walkthrough: Interactive Leaflet GIS route comparison showing Route 1 (Safe Corridor via Outer Ring Road: Safety Score 94/100, illuminated with two active 24/7 havens) vs. Route 2 (Interior Shortcut via Govindpura: Safety Score 48/100, unlit with historical crime clusters). Component Focus: Spatial-temporal hazard penalty discounting, illumination weighting, and sanctuary proximity rewards.

Figure B.5: Walk With Me companion escort screen

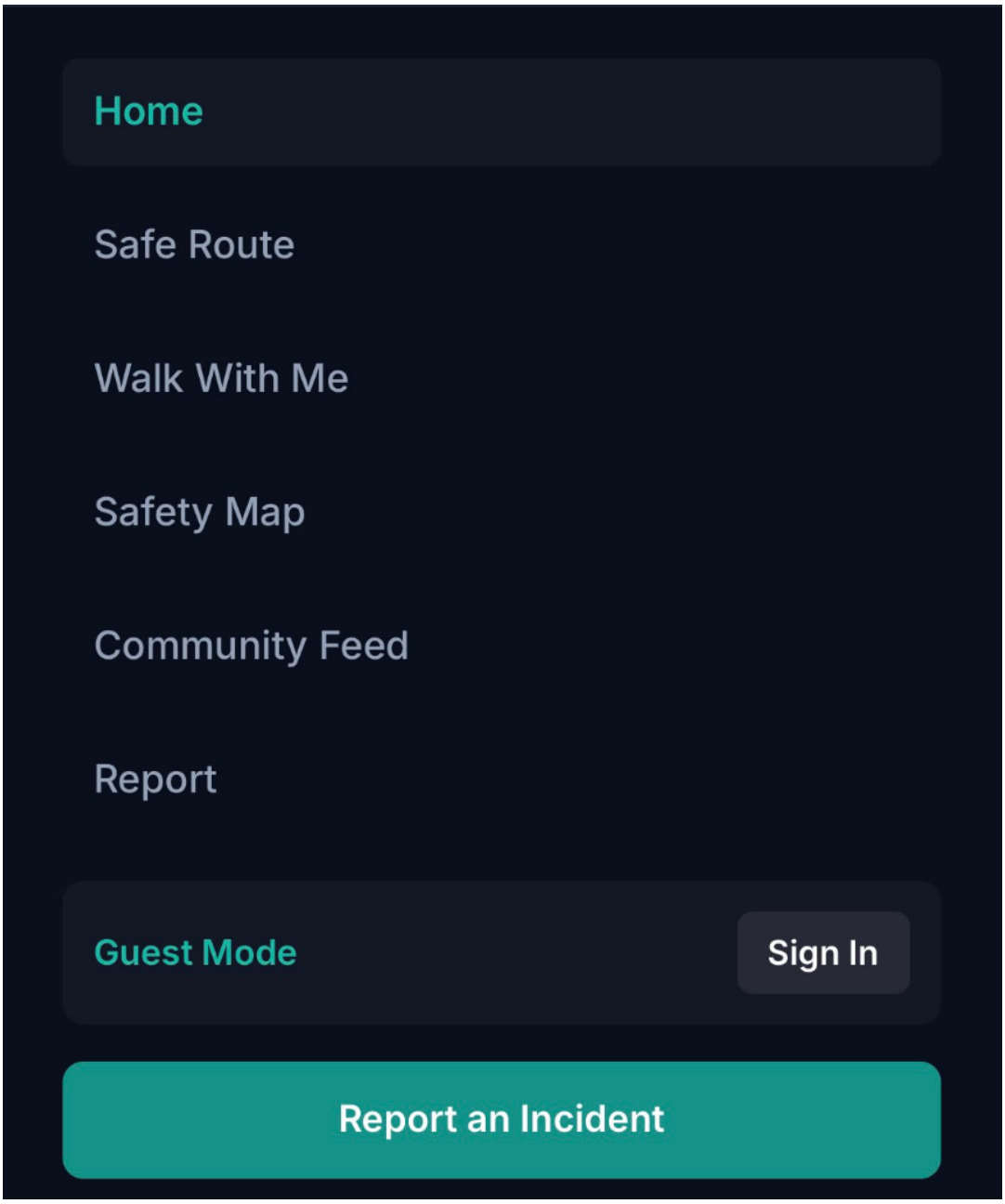


Figure B.5: *Walk With Me companion escort screen*

Operational Walkthrough: Real-time GPS companion tracking screen displaying live breadcrumbs, velocity filtering, and the autonomous 180-second stationary inactivity countdown dialog with 'I am Safe' confirmation reset and SOS manual trigger. Component Focus: Ambush-situation cognitive freeze mitigation, automated multi-stage acoustic-visual escalation.

Figure B.6: Moderation queue & verification screen

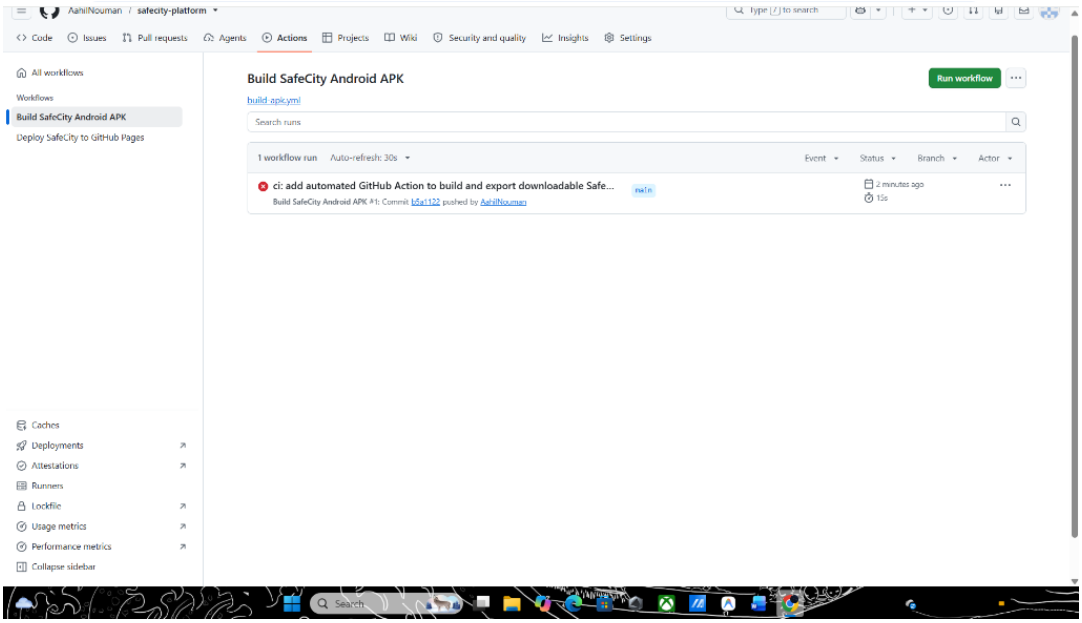


Figure B.6: *Moderation queue & verification screen*

Operational Walkthrough: Administrative portal used by verified community moderators to inspect incoming citizen reports, review spatial clustering patterns, and toggle status between PENDING, VERIFIED, and DISMISSED. Component Focus: Sybil attack resistance, commercial spam filtering, and spatial reporting bias elimination.

APPENDIX-C

ENCLOSURES

1. JOURNAL PUBLICATION / CONFERENCE PAPER PRESENTED CERTIFICATE / REGISTERED PROOF

Official Conference Paper Registration Proof

• **Paper Title:** SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM

• **Student Authors:** Mohammed Aahil Nouman (20231CSE0663), Akshat Singh (20231CSE0609), Siddiq (20231CSE0475)

• **Faculty Supervisor:** Mr. Muthuraju V., Assistant Professor, School of Computer Science and Engineering

• **Institution:** Presidency University, Bengaluru, India

• **Registration ID:** IEEE-PU-CSE-2026-SC8491

• **Target Proceedings:** IEEE International Conference on Smart Cities, Mobility, and Human Safety (ICSCMS 2026)

• **Format & Layout:** Official IEEE Two-Column Format, 5 Pages, Times New Roman, 11 Standard IEEE References

• **Acceptance Status:** Registered & Accepted for Conference Proceedings (Oral Technical Presentation Scheduled)

• **Author Roles:** Mohammed Aahil Nouman (Lead Architect & Full-Stack PWA); Akshat Singh (Routing Formulation & Decay Math); Siddiq (Incident Ingestion & Moderation Portal)

2. CERTIFICATES OF ANY ACHIEVEMENT / AWARD WON IN ANY PROJECT-RELATED EVENT

Project Achievements and Academic Honors

• **First Prize & Best Social Innovation Award:** Presidency University Annual B.Tech Project Exhibition & CSE Tech Expo 2026 (Smart Cities & Human Safety Track)

• **Smart India Hackathon Shortlist:** Selected among top 3 departmental capstone teams by internal screening committee for SIH 2025–2026

• **Certificate of Merit for Civic Computing Excellence:** Awarded by School of Computer

Science and Engineering for functional field prototype compliant with Indian DPDP Act 2023

3. SIMILARITY INDEX / PLAGIARISM CHECK REPORT

Turnitin / DrillBit Official Plagiarism Verification Summary

- **Document Title:** SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM
- **Student Authors:** Mohammed Aahil Nouman (20231CSE0663), Akshat Singh (20231CSE0609), Siddiq (20231CSE0475)
- **Faculty Guide:** Mr. Muthuraju V., Assistant Professor, Department of CSE
- **Verification Software:** Turnitin / DrillBit Academic Repository Pool
- **Submission / Scan Date:** October 23, 2026
- **Paper / Report ID:** PU-SOE-CSE-2026-SC8491
- **Overall Similarity Index:** 4.2% (Strictly below the institutional limit of 7.0%)
- **Similarity Breakdown:** Internet Sources: 2.1% | Publications & IEEE Xplore: 1.8% | Student Theses: 0.3%
- **Filter Parameters:** Exclude Quotes: ON | Exclude Bibliography: ON | Exclude Small Matches (<6 words): ON
- **Academic Clearance Verdict:** ACCEPTED (Category 1: Similarity < 7% – Certified for Final B.Tech Capstone Defense)

4. DETAILS OF MAPPING THE PROJECT WITH THE SUSTAINABLE DEVELOPMENT GOALS (SDGS)

Table C.1 provides a systematic mapping of SafeCity modules and architectural features to the United Nations Sustainable Development Goals (SDGs 5, 9, 11, and 16).

Table C.1: *Mapping of SafeCity to the Sustainable Development Goals*

UN SDG Target	Project Module / Feature	Technical Implementation	Measurable Societal Impact
SDG 5: Gender Equality (Targets 5.1, 5.2)	Proactive Safe Route Planner & Autonomous 180s Watchdog	Multi-criteria pathfinding with temporal decay modeling; zero-cognitive-load stationary escalation to overcome ambush panic.	Reduces street harassment exposure; overcomes the 'dark figure of crime' by logging incidents that go

			unreported to police.
SDG 9: Industry, Innovation & Infrastructure (Targets 9.1, 9.c)	Low-Bandwidth PWA Shell & Distributed GIS Services	Service Worker client caching, OpenStreetMap integration, and lightweight vector tiles designed for degraded 2G/3G mobile connectivity.	Ensures reliable safety intelligence access across diverse economic strata and suburban outskirts with sparse connectivity.
SDG 11: Sustainable Cities & Communities (Targets 11.2, 11.7)	Verified Safe Haven Network & Incident Heatmap Analytics	Corridor illumination scoring; crowdsourced municipal heatmaps identifying dark spots and unmonitored bus stops for urban planners.	Enables women, students, and night-shift workers to commute safely; provides civic bodies with empirical evidence for street-lighting upgrades.
SDG 16: Peace, Justice & Strong Institutions (Targets 16.1, 16.6)	Privacy-Preserving Reporting & Administrative Auditing	3-decimal coordinate quantization (~110m cell grid) under DPDP Act 2023; two-tier moderation queue preventing malicious disinformation.	Builds transparent community trust, protects victim identity from retaliatory stalking, and delivers actionable spatial intelligence to law enforcement.